

Zihan Liang

PHD STUDENT · COMPUTATIONAL BIOLOGY AND BIOINFORMATICS PROGRAM, DUKE

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Education

Duke University

Durham, NC

PH.D. IN COMPUTATIONAL BIOLOGY AND BIOINFORMATICS

Aug. 2026 - Present

- Ph.D. Advisor: [Dr. Anru Zhang](#)

Emory University

Atlanta, GA

BACHELOR OF SCIENCE IN APPLIED MATHEMATICS AND STATISTICS WITH HIGHEST HONORS

Aug. 2023 - May 2026

- Minors in Artificial Intelligence
- Independent Research and Honors Program Advisor: [Dr. Ruoxuan Xiong](#)
- Relevant Courseworks:
 - Mathematics: Multivariable Calculus, Linear Algebra, Discrete Mathematics, Differential Equations, Linear Optimization, Graph Theory (Graduate Level), Real Analysis
 - Statistics & Computation: Mathematical Statistics, Numerical Analysis, Data Structures & Algorithms, Data Science Computing, Machine Learning, Text Data Science, Information Visualization, Experimental Methods, Causal Inference (Doctoral Level)

Publications

PUBLISHED

- Pan, Z^{*}, **Liang, Z^{*}**, Xiong, R. 2026. Two-Sided State-Space Models for Sequential Recommendation with Non-Random Multimodal Review Feedback. Findings of the Association for Computational Linguistics: EMNLP 2026.
- Pan, Z^{*}, **Liang, Z^{*}**, Kabbara, J., Emami, A. 2026. DART: Mitigating Harm Drift in Difference-Aware LLMs via Distill-Audit-Repair Training. Findings of the Association for Computational Linguistics: ACL 2026.
- Liang, Z^{*}**, Pan, Z^{*}, Xiong, R. 2026. Learning Dynamic Representations and Policies from Multimodal Clinical Time-Series with Informative Missingness. Findings of the Association for Computational Linguistics: ACL 2026.
- Shen, Y^{*}, Jiang, D^{*}, **Liang, Z^{*}**, Wei, Y^{*}, Yang, W., Sun, X., Huang, Z., Wang, S., Yao, Q. 2026. MambaDATG: Domain-Adaptive Tri-Plane-Gated Pre-training for 3D Abdominal Segmentation. IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP 2026), Oral Presentation.
- Liang, Z^{*}**, Pan, Z^{*}, Xiong, R. 2025. Causal Representation Learning from Multimodal Clinical Records under Non-Random Modality Missingness. Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP 2025).
- Liang, Z^{*}**, Pan, Z^{*}, Dey, S. K., Ismail, A. 2025. CareLab at #SMM4H-HeaRD 2025: Insomnia Detection and Food Safety Event Extraction with Domain-Aware Transformers. Proceedings of the International AAAI Conference on Web and Social Media (ICWSM 2025) Workshops.
- Liang, Z.**, Pan, Z. 2026. Dynamic Policy Design for Autonomous Taxi Adoption: A Hierarchical Game-Theoretic Framework. Transportation Research Board 105th Annual Meeting (TRB 2026).

IN REVIEW

- Liang, Z^{*}**, Pan, Z^{*}, Yu, S. 2026. Adaptive Spatiotemporal Graph Neural Networks with Trend-Aware Prediction and Validation-Gated Calibration for Traffic Flow Forecasting. IEEE Transactions on Intelligent Transportation Systems, under review.

Research Experience

Collective Action & Research for Equity (CARE) Lab, Emory University

Atlanta, GA

RESEARCH ASSISTANT, ADVISOR: DR. AZRA ISMAIL

Sept. 2024 - May 2026

- Built an interactive dashboard for nearly 1,000 user records using JavaScript, Python, and HTML.
- Maintained a WhatsApp education bot serving 15,000+ teachers and 600,000+ students across India.
- Developed an offline-first Android app for classroom data collection and secure Google Drive synchronization.

Polymath Jr REU Summer Program

Remote

RESEARCH ASSISTANT

Jun. 2025 - Aug. 2025

- Developed ODE-based models to study community transmission of *C. difficile*.
- Evaluated intervention strategies for controlling community spread of *C. difficile*.
- Implemented disease simulations and mitigation analyses in MATLAB and R.

Honors & Awards

2026 Elected to Phi Beta Kappa (ΦBK), Phi Beta Kappa Society

2025 Best Visualization Award, American Statistical Association DataFest

2025 Conference Presentation Grant, Emory University

\$2,500

2023, 2024 Dean's List, Emory College of Arts and Sciences

2022 American Invitational Mathematics Examination (AIME), Score: 9/15, Mathematical Association of America

Teaching Experience

2026 Teaching Assistant and Grader of Machine Learning, Emory University

Atlanta, GA

2025 Teaching Assistant and Grader of Algebra for Mathematical Olympiads, AwesomeMath

Remote

Outreach, Professional Development & Leadership Experiences

LEADERSHIP EXPERIENCES

2023-2026 Emory Chinese Student Association, President

Atlanta, GA

2024-2026 Emory Artificial Intelligence & Data Association, Founder and President

Atlanta, GA

2022-2024 YouthBuild Program - Xi'an, Co-Founder & Standing Council Member

Xi'an, Shaanxi

PEER REVIEW

Reviewer, ACL 2026 Workshop on Social Media Mining for Health and Health Real-World Data (SMM4H-HeaRD)

Reviewer, ACM CIKM 2026 Workshop on LLM Agents for Social Simulation (LASS)